# Solutions and teaching notes, chapter 6

**Teacher use.** The following answers concern the tasks not solved in the lesson. First have pupils make their approach explicit; provide graduated help before the solution. The durations are estimates to be calibrated in class.

## Diagnostic and remediation

1. 6. *Action:* Go back to the rate 2 $a+h$.
2. $\mathbb{R}\setminus\{2\}$. *Action:* Set the denominator to zero.
3. Positive outside [ $-2$; 1], negative between the roots, zero at the endpoints. *Action:* Construct two sign rows.
4. $x^3-2x^2+x-2$. *Action:* Distribute each term without leaving any out.
5. They are equal. *Action:* Substitute -x in brackets.
6. $y=2$. *Action:* Go back to the tangent equation and simplify.

## Investigation

$x^3$ is strictly increasing with a horizontal tangent at 0. For a differentiable increasing function, the rates on both sides are positive or zero, so the derivative cannot be negative.

## Direct applications

1. $5x^4$ on $\mathbb{R}$ ; $-\dfrac{2}{x^3}$ for $x\neq 0$ ; $\dfrac{1}{2\sqrt{x}}$ for $x>0$.
2. $2x(3x-2)+3(x^2+1)=9x^2-4x+3$ ; $-\dfrac{2x}{(x^2+1)^2}$ on $\mathbb{R}$.
3. $f\prime=2x-4$, so decreasing up to 2, then increasing. Minimum $-3$ at 2; values at the endpoints 1 and $-2$, so the maximum is 1 at 0.
4. First even, second odd; difference $x^2+1-2x=(x-1)^2\geq 0$, equality at 1.
5. $A(x)=x(6-x)$ on [0; 6]; $A\prime=6-2x$. Maximum 9 m $^2$ at 3.

## Exercises: methods and answers

**Exercise 6.1 · A collection of derivatives.** $12x^3-2$ ; $2x(x-1)+(x^2+2)=3x^2-2x+2$ ; $\dfrac{[2(x+3)-(2x+1)]}{(x+3)^2}=\dfrac{5}{(x+3)^2}$ for $x\neq -3$.

**Exercise 6.2 - A complete table.** $f\prime=3(x-1)(x-3)$ ; signs + , −, +. Values $f(0)=1$, $f(1)=5$, $f(3)=1$, $f(4)=5$. Minimum 1 at 0 and 3; maximum 5 at 1 and 4. The endpoints count.

**Exercise 6.3 - A quotient.** $f\prime=1-\dfrac{4}{x^2}=\dfrac{(x-2)(x+2)}{x^2}$. Negative before 2, positive after. Minimum 4 at 2; for $x>0$, $x+\dfrac{4}{x}\geq 4$, equality only at 2.

**Exercise 6.4 - Comparing two curves.** $f - g = x(x-1)(x+1)$. Signs: negative on $]-\infty; -1[$, positive on $]-1; 0[$, negative on $]0; 1[$, positive on $]1; +\infty[$. Intersections at $-1, 0, 1$. We can draw up the table of the three factors without differentiation.

**Exercise 6.5 - A box.** $V = x(12-2x)^2 = 4x(6-x)^2$. $V' = 12(x-2)(x-6)$, positive on $[0; 2[$, negative on $]2; 6[$. $V(0) = V(6) = 0$, $V(2) = 128$ cm $^3$. Maximum at x=2 cm; box 8 $\times 8 \times 2$.

**Exercise 6.6 - Stationary point without an extremum.** $f' = 3x^2$, so it is 0 at 0. In every neighbourhood of 0, negative x values give $f(x) < 0$ and positive x $f(x) > 0$; no extremum.

**Exercise 6.7 - Comparing strategies.** Square: $(x-3)^2 + 3 \geq 3$. Derivative: $2x - 6$, minimum at 3, value 3. The first avoids a complete study of variations; the second illustrates the link between chapters.

**Exercise 6.8 - Delayed review.** It decreases on $]-\infty; 0[$ and $]0; +\infty[$. It does not decrease on their union: $-1 < 1$ but $f(-1) = -1 < 1 = f(1)$. The domain is not an interval.

**CQFD, exercise 12 p. 156, questions 1 and 2.** $f(x) - g(x) = \frac{1}{2}x^2 - 1 = \frac{1}{2}(x-\sqrt{2})(x+\sqrt{2})$. The curve of f lies strictly above that of g for $x < -\sqrt{2}$ or $x > \sqrt{2}$, below between these two numbers; equality at the x-coordinates $\pm\sqrt{2}$.

**CQFD, exercise 16 p. 156, questions f and g.** For $f(x) = x^3$, we can choose $]-\infty; 0[$ : $f(x) < 0$ and the function is strictly increasing. For $g(x) = x^2 - 4$, we can choose $[0; 2[$ : $g(x) < 0$ and the function is strictly increasing on this interval. Subintervals not reduced to a single point are also suitable.

## MCQ: answers and reasons

**MCQ 1.**

1. B. Integer power.

2. A. Rate of change of the reciprocal.

3. D. Product rule.

4. C. Sign of the derivative and variation.

5. B. Non-zero denominator.

**MCQ 2.**

1. C. $f(-x) = -f(x)$.

2. A. The other conclusions need more.

3. D. Formula and domain.

4. B. Attained at 1.

5. C. Derivative $2x_0 = 0$.

## True or false

1. False: $f(x) = \dfrac{1}{x} > 0$ on $]0; +\infty[$, but $f\prime(x) < 0$.

2. True, with the stated assumptions; at the endpoints, the conclusion is not forced.

3. False without sign assumptions: on $[-2; -1]$, $u(x) = v(x) = x$ increase but $x^2$ decreases.

**Differentiated teaching.** If a pupil is stuck on a technical point, suggest only the first step (domain, factor or reference sample space), then remove the help. For an independent student, ask for a second method and the handling of a boundary case. Orally, allow an initial rephrasing in the language the pupil is fluent in, then require a precise mathematical sentence in French.